

# Shreyash Bhatkar

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## EDUCATION

**University of Illinois Urbana-Champaign** | Master of Computer Science, GPA: **4.0/4.0** Expected 2027

**University of Pune** | B.Tech Electronics and Computer Science, CGPA: **3.62/4.0** Aug 2019 – May 2023

## TECHNICAL SKILLS

**Programming:** Java, Python, C++, Go, Rust, TypeScript, JavaScript, SQL, Bash

**Software Engineering:** Spring Boot, Node.js, React.js, REST APIs, gRPC, Microservices, Distributed Systems, Concurrency, DSA, OOP

**Cloud & Data:** AWS, Docker, Kubernetes, Kafka, Redis, PostgreSQL, Cassandra, JPA, CI/CD, Linux, Slurm/HPC

**AI & Developer Tools:** PyTorch, vLLM, RAG, Git/GitHub, Jenkins, Jest, Prometheus, OpenTelemetry, MLflow

## EXPERIENCE

**Oracle Corporation** Jul 2023 – Jan 2026

*Software Engineer | Client Engagements: Batelco Telecom & PPC Retail India*

- Built 8 Java/Kotlin Spring Boot microservices and 20+ REST APIs handling 10M+ daily transactions; saga recovery and Redis idempotency maintained 99.95% correctness under concurrent load.
- Traced gRPC HTTP/2 stream exhaustion with Prometheus and tuned Istio/OCNE connection pools, increasing throughput by 37% and reducing P95 latency by 12%.
- Built Kafka event pipelines with retry and idempotent processing across 12+ integrations, raising throughput by 35% with zero duplicate-charge incidents after rollout.
- Optimized SQL/JPA queries and indexes over 300M+ monthly records, cutting peak-cycle database latency by 20%.
- Delivered 13+ JavaScript/Node.js product features and reusable APIs, reducing page-load time by 23% and backend load by 17%.
- Led a 4-person team building a production RAG assistant for 500+ engineers and 2K+ daily queries, reducing hallucinations by 41% and ticket-resolution time by 33%.

**SSAIL – Supercomputing Systems & AI Lab, UIUC** Jan 2026 – Present

*Graduate Researcher – Systems & Runtime Engineering | Advisor: Prof. Minjia Zhang Champaign, IL*

- Built an NCSA DeltaAI-to-AWS execution path for Dockerized SWE-Bench workloads, validating reproducible orchestration across HPC and cloud infrastructure.
- Profiled TRL/TorchForge across latency, memory, utilization, and placement; found asymmetric GPU allocation behind an apparent speedup and redesigned matched experiments.
- Debugged FSDP2/vLLM/NCCL synchronization failures on 4× RTX PRO 6000 GPUs using source tracing, controlled A/B tests, and process-level telemetry.

**University of Illinois Urbana-Champaign** Jun 2026 – Present

*Graduate Research Assistant – Human-Robot Systems Champaign, IL*

- Operated a 6-DOF manipulator and built RGB-D/3D data pipelines with <50ms event latency; calibrated camera-to-robot mappings using SE(3) transforms.
- Built Grasshopper components that convert parametric 3D geometry into machine-readable control data for robot sequencing and fabrication workflows.

**Nibha Tech Solutions** Dec 2022 – May 2023

*AI Research Intern – Financial Markets Nashik, India*

- Developed C++/Python feature-processing pipelines over 25 years of market data, generating 30,000+ research features for quantitative modeling.
- Integrated 15GB+ datasets into backend/ML pipelines and evaluated SARIMA/SHAP models, reaching 96.2% accuracy in normal-market classification.

## PROJECTS

**Synchronized Code Editor** | TypeScript, React.js, Node.js, Socket.io, WebRTC, Redis, Docker, AWS, Jest

- Built a real-time collaborative editor using operational transform and WebSockets, sustaining <100ms synchronization latency across concurrent editing sessions.
- Deployed Redis-backed services on AWS and added Jest integration tests, supporting 300+ concurrent users at 99.9% uptime.

**Scalable Distributed File System (ScDFS)** | C++, Linux, Cassandra, PostgreSQL, Multithreading

- Implemented consistent hashing, Cassandra metadata, 3× replication, and node recovery in C++; multithreaded sharding reduced query latency by 40%.

## AWARDS & CERTIFICATIONS

- Oracle Above and Beyond Award | Pat on the Back Award | Runner-Up, CGIU Memorial AI Hackathon | Oracle Generative AI & Data Science Professional | Indian Provisional Patent – Edge AI Safety Wearable